

A PROJECT TITLE
ON
CUSTOMER PURCHASE PREDICTION USING LOGISTIC
REGRESSION AND RANDOM FOREST

project report submitted in partial fulfilment of the requirements for the Course

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Abstract

Customer Purchase Prediction Using Logistic Regression and Random Forest is a machine-learning project developed using Python to predict whether a customer is likely to make a purchase. The system uses demographic and behavioural features including age, gender, annual income, previous purchases, website visits, pages viewed, time spent on the website, emails opened, cart additions, and days since the last purchase. The dataset initially contains 5,000 customer records and 13 columns. Data preprocessing includes duplicate removal, identifier removal, missing-value treatment, categorical encoding, and numerical scaling. Logistic Regression and Random Forest are trained using leakage-safe pipelines. A tuned Random Forest is also evaluated using 5-fold GridSearchCV. The models are compared using accuracy, precision, recall, F1-score, and ROC-AUC. On the held-out test set, Logistic Regression achieved 61.77% accuracy and a 41.36% F1-score, while Random Forest achieved 57.95% accuracy and a 34.69% F1-score. Unsupervised learning techniques including K-Means clustering, the Elbow Method, Silhouette Score, DBSCAN, and Hierarchical Clustering were also applied to explore customer groups. The project demonstrates a complete machine-learning workflow combining supervised purchase prediction and unsupervised customer clustering.

Keywords: customer purchase prediction; Logistic Regression; Random Forest; classification; customer behavior; machine learning.

1. Introduction

1.1 Purpose of the Project

Customer purchase prediction is a machine-learning application used to estimate whether a customer is likely to purchase a product or service. Such predictions can support customer targeting, campaign planning and prioritization of potential buyers.

1.2 Problem Statement

The problem is a binary classification task. Given customer demographic and behavioural information, the system predicts the target variable Purchase, where 0 represents no purchase and 1 represents a purchase.

1.3 Existing Methods and Disadvantages

Traditional customer targeting may rely on manual rules or basic segmentation. These approaches may not capture interactions among several customer characteristics and behavioural variables. A machine-learning classifier can learn patterns from historical customer records and provide a consistent prediction.

1.4 Proposed System

The proposed system loads the customer dataset, checks data quality, removes duplicate records and the Customer_ID identifier, handles missing values, performs exploratory data analysis, applies unsupervised clustering techniques for customer grouping, creates leakage-safe preprocessing pipelines, trains Logistic Regression and Random Forest models, tunes Random Forest hyperparameters, evaluates the models using multiple metrics, and demonstrates a final prediction for a new customer.

1.5 Research Objectives

- Prepare and describe the customer purchase dataset.
- Perform systematic preprocessing and exploratory data analysis.
- Apply K-Means, Elbow Method, Silhouette Score, DBSCAN, and Hierarchical Clustering to explore customer groups.
- Remove identifier information and prevent data leakage.
- Develop and train Logistic Regression and Random Forest classifiers.
- Perform hyperparameter tuning and compare model performance.
- Demonstrate purchase prediction on a new customer record.

2. Literature Survey

The literature review was expanded to ten studies directly related to customer purchase prediction, online purchase behaviour, e-commerce clickstream analysis and machine-learning classification. The methodology and findings are summarized below. The table follows the same structure as the reference report.

S.No.	Title of the paper	Authors / Journal & Year	Methodology Adapted	Key Findings	Gaps
1	Modeling Purchase Behavior at an E-Commerce Web Site: A Task-Completion Approach	Catarina Sismeiro, Randolph E. Bucklin — Journal of Marketing Research, 2004	Sequential task-completion modelling using clickstream data and binary probit models.	Browsing and navigational behaviour can predict task completion and help identify likely buyers early.	Uses a specific automotive website and sequential task framework; not directly a general tabular customer classifier.
2	Purchase Prediction Using Tmall-Specific Features	Y. Zhao, L. Yao, Y. Zhang — Concurrency and Computation: Practice and Experience, 2016	Extracted click, purchase, collect and cart feature groups from Tmall activity; threshold-moving for prediction.	Combining the three feature groups provided useful purchase-prediction information; collect-and-cart features were particularly important.	Tmall-specific features and user-brand prediction limit direct transfer to small customer-level datasets.
3	A Machine Learning Framework for Customer Purchase Prediction in the Non-Contractual Setting	A. Martínez, C. Schmuck, S. Pereverzyev Jr., C. Pirker, M. Haltmeier — European Journal of Operational Research, 2020	Dynamic customer features from previous purchases; Logistic Lasso, extreme learning machine and gradient tree boosting.	Gradient tree boosting achieved 89% accuracy and 0.95 AUC for next-month purchase prediction on the studied dataset.	Requires longitudinal transaction history and a defined future purchase window.
4	Towards Early Purchase Intention Prediction in Online Session Based Retailing Systems	Ramazan Esmeli, Mohamed Bader-El-Den, Hassana Abdullahi — Electronic Markets, 2021	Session-based online behavioural features for early purchase-intention prediction.	Shows the value of predicting purchase intention before the end of a session for e-commerce decision support.	Session-level setting differs from the demographic and customer-history structure of this project.
5	On the Platform but Will They Buy? Predicting Customers' Purchase Behavior Using Deep Learning	Neha Chaudhuri, et al. — Decision Support Systems, 2021	Deep neural networks using platform engagement and customer characteristics; compared with Decision Tree, Random Forest, SVM and ANN.	Deep learning outperformed the compared conventional ML methods on the studied e-commerce dataset and highlighted platform/customer predictors.	More complex deep-learning architecture and large session dataset are beyond the scope of this compact project.
6	Customer Purchase Prediction from the Perspective of Imbalanced Data: A Machine Learning Framework Based on Factorization Machine	Research article — Expert Systems with Applications, 2021	Improved factorization-machine feature combinations with Random Forest and cost-sensitive learning for imbalanced purchases.	Addresses sparse behavioural features and class imbalance through feature combination and cost-sensitive learning.	The proposed framework is more complex than the two-model baseline used in this project.
7	Online Purchase Prediction via Multi-Scale Modeling of Behavior Dynamics	Chao Huang, Xian Wu, Xuchao Zhang, Chuxu Zhang, Jiashu Zhao, Dawei Yin, Nitesh Chawla — ACM SIGKDD, 2019	Graph Multi-Scale Pyramid Networks with convolution-recurrent modelling of temporal and category dependencies.	Models complex temporal behaviour and category dependencies for online purchase forecasting.	Requires sequential interaction data and deep neural architectures rather than compact customer-level tabular data.
8	Research on E-Commerce Purchase Prediction Model Based on Machine Learning	Zaixin Lin, Zihui Huang, Huiyan He, Jiazi Liang, Xinbei Zheng, Jingde	Random Forest decision model using features from customer historical	Random Forest was reported as a practical model for predicting	The study focuses on Random Forest and does not provide the same

		Huang — 2023	purchase data.	customer purchasing behaviour.	Logistic Regression versus Random Forest comparison used here.
9	E-Commerce User Purchase Prediction Based on Improved Machine Learning Algorithms	IEEE SMART GENCON, 2023	Markov-chain-based purchase-behaviour evaluation combined with machine-learning analysis; addresses incomplete and imbalanced data.	Improved machine-learning analysis was reported to outperform the referenced k-NN approach under the study's evaluation criteria.	Uses a different modelling framework and simulation setting from this project's customer-level dataset.
10	Predicting Online Customer Purchase: The Integration of Customer Characteristics and Browsing Patterns	Decision Support Systems, 2024	Combines RFM customer characteristics with browsing-pattern and clickstream information for session-level prediction and segmentation.	Integrating customer characteristics and browsing patterns can improve purchase prediction and support personalized marketing.	Requires richer clickstream and transaction histories than the available 500-record dataset.

2.1 Literature Review Summary

The reviewed studies show three recurring directions: (1) customer and browsing features are useful for purchase prediction, (2) tree-based and ensemble methods can model nonlinear customer behavior, and (3) richer sequential or clickstream data can improve prediction but also increases modelling complexity. The present project focuses on a compact, interpretable supervised-learning workflow using Logistic Regression and Random Forest with demographic and behavioral customer features.

3. Methodology

3.1 Proposed System Flow Diagram

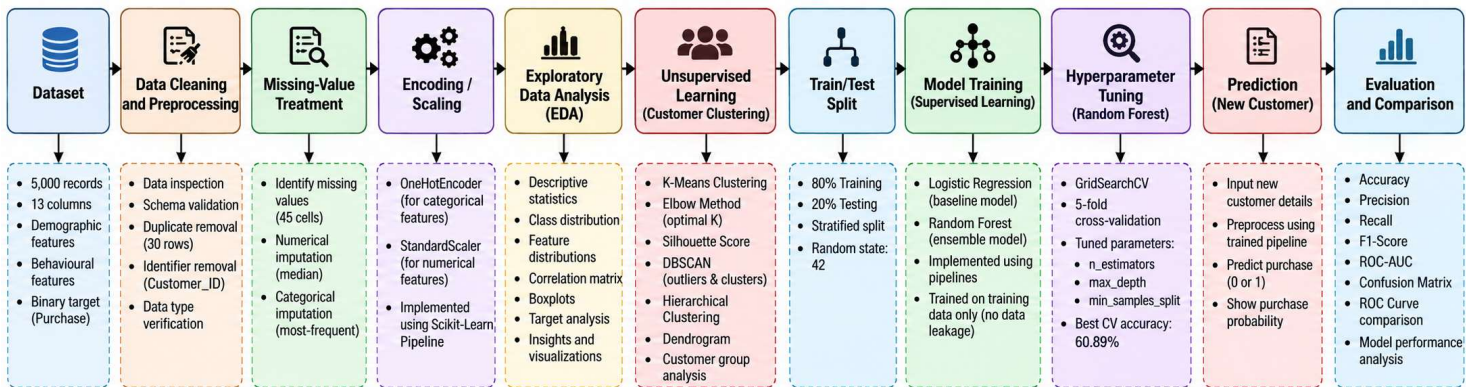


Figure 1. Proposed System Flow Diagram.

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3.2 Dataset Description

The dataset contains **5,000 customer records and 13 columns**. It includes customer demographic information, income, purchase history, website engagement and the binary Purchase target. The dataset contains 30 duplicate rows and 45 missing cells. Customer_ID is an identifier and is removed before model training.

Property	Value
Records	5000
Total columns	13
Target	Purchase
Original class distribution	2,921 No Purchase; 2,079 Purchase
Duplicate rows	30
After duplicate removal	4,970 rows

Missing values after duplicate removal	45 cells
Modelling predictors	4,970 rows with 11 predictors and 1 target after removing Customer_ID

3.3 Experimental Setup

Component	Configuration
Environment	Jupyter Notebook
Operating system	Windows
Python	Python 3.x
Libraries	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, SciPy
Train/Test split	80% training, 20% testing; stratified
Random state	42

3.4 Data Preprocessing

- Data inspection, schema validation, missing-value checking and duplicate checking were performed.
- 30 duplicate rows were identified and removed, leaving 4,970 unique records.
- 45 missing cells were identified in the original dataset.
- Customer_ID was removed because it is an identifier rather than a behavioural predictor.
- Numerical missing values were handled using median imputation.
- Categorical missing values in Gender and Customer_Segment were handled using most-frequent imputation.
- Categorical variables were converted using OneHotEncoder.
- Numerical variables were standardized using StandardScaler.
- The preprocessing operations were placed inside the Scikit-Learn pipeline so that preprocessing parameters were learned from the training data rather than the test data.

3.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the structure, distribution, and relationships within the customer dataset. The analysis included descriptive statistics, target-variable distribution, numerical feature distributions, categorical feature distributions, boxplots across purchase categories, and a correlation heatmap.

The original dataset contained **5,000 customer records**, with **2,905 customers classified as No Purchase** and **2,095 customers classified as Purchase**. Duplicate records were subsequently removed, leaving **4,970 unique records** for further analysis and modelling.

Numerical features such as **Age, Annual_Income, Previous_Purchases, Website_Visits, Pages_Viewed, Time_on_Website_Min, Emails_Opened, Cart_Additions, and Days_Since_Last_Purchase** were examined using descriptive statistics and distribution plots. Categorical variables such as **Gender and Customer_Segment** were also analysed to understand their distributions.

Boxplots were used to compare numerical customer characteristics across the purchase classes. A correlation heatmap was used to examine linear relationships among the numerical features and the target variable **Purchase**. The observed correlations were generally weak, indicating that no single numerical

feature had a strong linear relationship with the purchase outcome. Therefore, multiple customer demographic and behavioural features were considered together during model development.

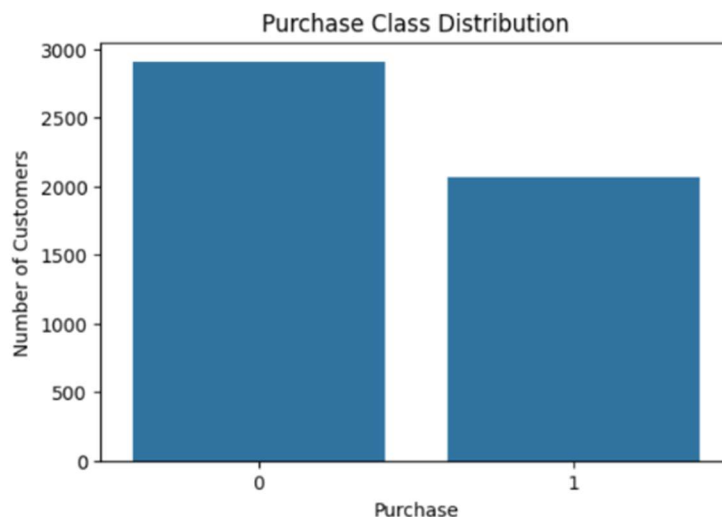


Figure 2. Purchase-class distribution in the original dataset.

3.6 Feature Engineering and Feature Selection

No artificial features were required for the final model. Customer_ID was excluded because it is an identifier. The final predictor set consisted of Age, Gender, Annual_Income, Customer_Segment, Previous_Purchases, Website_Visits, Pages_Viewed, Time_on_Website_Min, Emails_Opened, Cart_Additions and Days_Since_Last_Purchase. Purchase was used as the target.

3.7 Unsupervised Learning

Unsupervised learning was performed to identify natural customer groups without using the target variable Purchase. Numerical customer features were median-imputed and standardized before clustering.

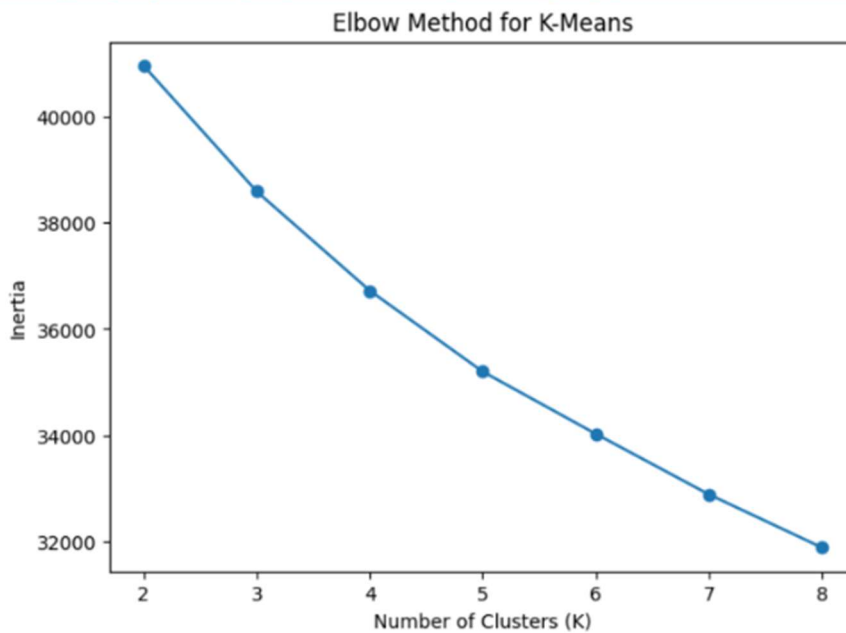
3.7.1 K-Means Clustering

K-Means clustering was applied to standardized numerical customer features to identify groups of customers with similar characteristics.



3.7.2 Elbow Method

The Elbow Method was used to evaluate different values of K by examining the K-Means inertia curve. K values from 2 to 8 were tested.



3.7.3 Silhouette Score

The Silhouette Score was calculated for K values from 2 to 8 to evaluate cluster separation and cohesion. The highest tested Silhouette Score was approximately 0.0837 at K=2.

```
K-Means (k=3) Silhouette Score: 0.0737
DBSCAN Detected Clusters: 32
DBSCAN Detected Outliers (noise): 1353
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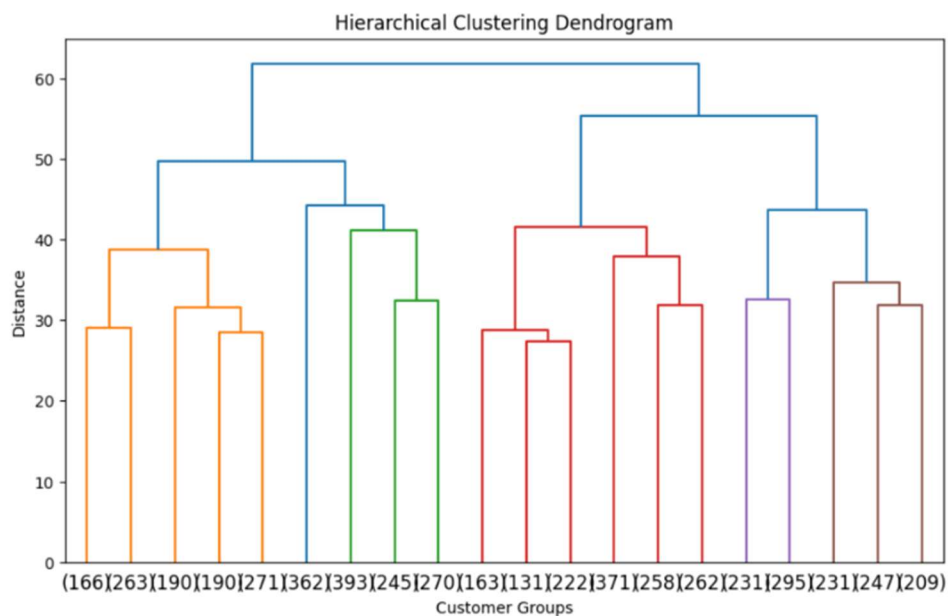
3.7.4 DBSCAN

DBSCAN was applied using $\text{eps} = 1.5$ and $\text{min_samples} = 5$. The model identified 32 clusters and 1,353 observations as noise or outliers.

```
K-Means (k=3) Silhouette Score: 0.0737
DBSCAN Detected Clusters: 32
DBSCAN Detected Outliers (noise): 1353
```

3.7.5 Hierarchical Clustering and Dendrogram

Hierarchical clustering was performed using Ward's linkage. A dendrogram was generated to visualize the hierarchical merging of customer groups.



3.8 Model Building and Training

Model	Purpose	Configuration
Logistic Regression	Linear baseline for binary classification	max_iter=1000; preprocessing pipeline; standardized numerical variables; one-hot categorical variables
Random Forest	Nonlinear ensemble classifier	n_estimators=100; random_state=42; preprocessing pipeline

3.9 Hyperparameter Tuning

GridSearchCV with 5-fold cross-validation was applied to the Random Forest pipeline. The search considered n_estimators, max_depth and min_samples_split. Accuracy was used as the cross-validation scoring metric.

Tuning Item	Value
Search method	GridSearchCV
Cross-validation	5-fold
Scoring metric	Accuracy
Search parameters	n_estimators, max_depth, min_samples_split
Best parameters	n_estimators=50, max_depth=10, min_samples_split=5
Best CV accuracy	60.89%

3.10 Model Evaluation

The test set was held out from model fitting. Accuracy, precision, recall, F1-score, ROC-AUC, classification reports and confusion matrices were used. ROC-AUC was calculated using predicted probabilities.

4. Results and Discussion

4.1 Preprocessing Results

Stage	Result
Initial dataset	5000 rows and 13 columns
Duplicate check	30 duplicate rows removed
After duplicate removal	4,970 rows
Identifier removal	Customer ID removed
Missing-value treatment	Missing-value treatment 45 missing cells; median imputation for numerical; most-frequent for categorical
Final modelling data	4970 rows with 11 predictors and 1 target

4.2 Model Performance

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	61.77%	57.02%	32.45%	41.36%	0.6351
Random Forest	57.95%	48.90%	26.88%	34.69%	0.5775
Tuned Random Forest	59.86%	53.40%	26.63%	35.54%	0.6046

Logistic Regression achieved the highest held-out test accuracy and F1-score among the three models. Random Forest achieved the highest ROC-AUC. The tuned Random Forest improved its cross-validation accuracy during model selection, but its held-out test performance was lower than the original Random Forest on this split.

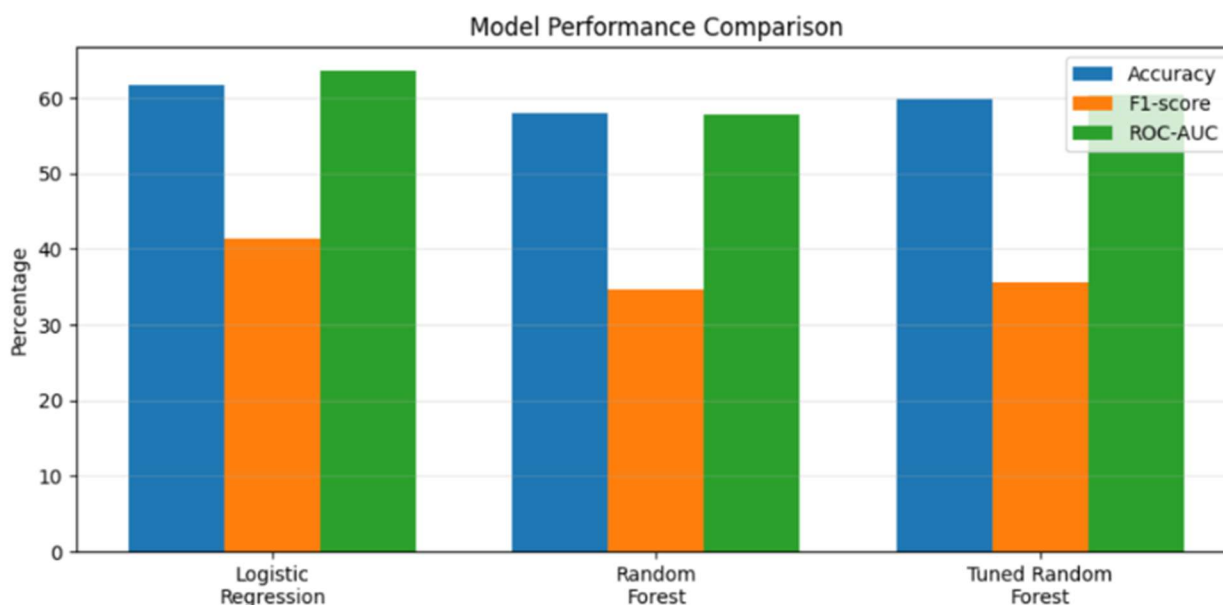


Figure 3. Model performance comparison.

4.3 Confusion Matrix Results

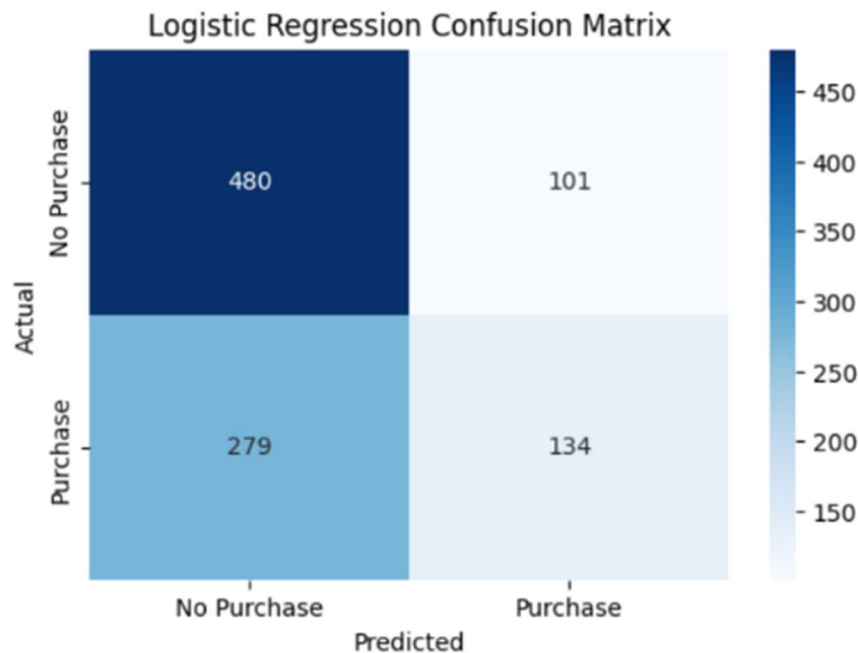


Figure 4. Confusion matrix for Logistic Regression.

The Logistic Regression confusion matrix was $[[480, 101], [279, 134]]$, corresponding to 480 true negatives, 101 false positives, 279 false negatives, and 134 true positives. The model identified no-purchase customers more effectively than purchase customers, as reflected by the higher number of correctly classified no-purchase customers.

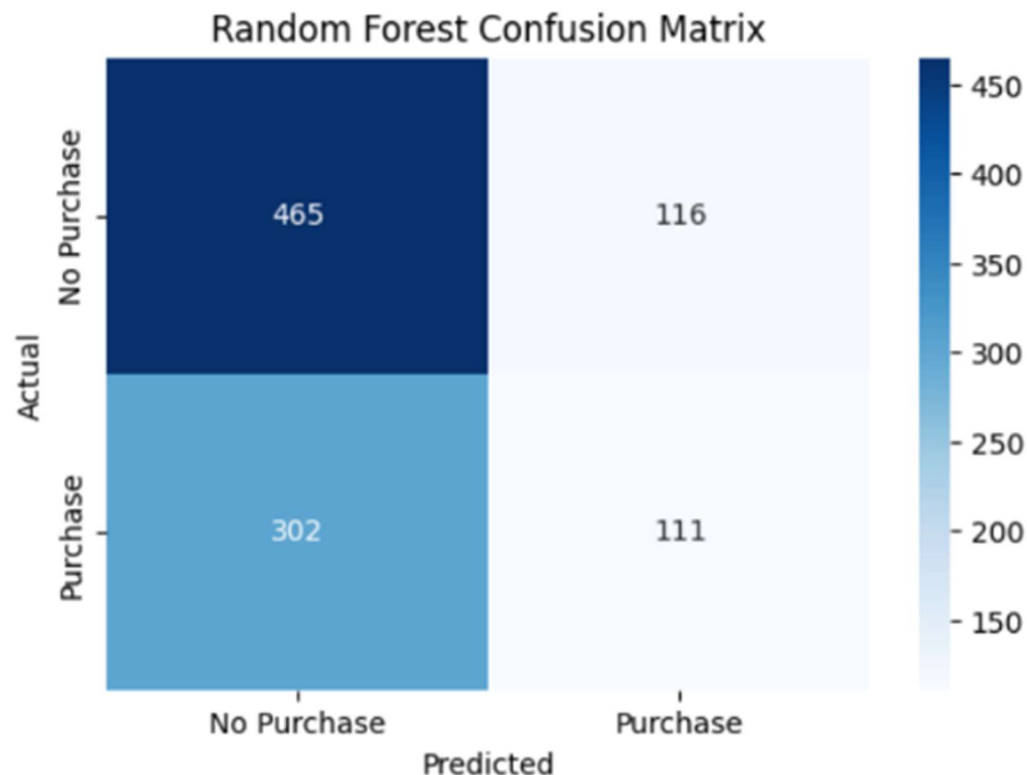


Figure 5. Confusion matrix for Random Forest.

The Random Forest confusion matrix was $[[465, 116], [302, 111]]$, corresponding to 465 true negatives, 116 false positives, 302 false negatives, and 111 true positives. The model classified more no-purchase customers correctly than purchase customers.

4.4 Random Forest Feature Importance

Feature	Importance
Days_Since_Last_Purchase	0.1677
Annual_Income	0.1409
Time_on_Website_Min	0.1090
Website_Visits	0.1028
Age	0.1024
Emails_Opened	0.0861
Pages_Viewed	0.0821
Previous_Purchases	0.0697
Cart_Additions	0.0657

The fitted Random Forest assigned relatively higher importance to Days_Since_Last_Purchase, Annual_Income, Time_on_Website_Min, Website_Visits and Age. Feature importance indicates contribution to model predictions and does not establish a causal relationship.

4.5 Results and Discussion

Three classification models were implemented and evaluated: Logistic Regression, Random Forest and Tuned Random Forest. Logistic Regression achieved 61.77% accuracy, 57.02% precision, 32.45% recall, 41.36% F1-score and 0.6351 ROC-AUC. Random Forest achieved 57.95% accuracy, 48.90% precision, 26.88% recall, 34.69% F1-score and 0.5775 ROC-AUC. The tuned Random Forest achieved 60.89% cross-validation accuracy and 59.86% held-out test accuracy, with 53.40% precision, 26.63% recall, 35.54% F1-score and 0.6046 ROC-AUC. The results indicate that the models classified the no-purchase class more effectively than the purchase class, as reflected by the relatively lower recall for the purchase class.

5. Final Prediction / Demonstration

Input Feature	Value
Age	35
Gender	Male
Annual Income	80000
Customer Segment	Regular
Previous Purchases	5
Website Visits	10
Pages Viewed	6
Time on Website (min)	12
Emails Opened	4
Cart Additions	2
Days Since Last Purchase	60

Predicted Purchase: 0 (No Purchase) , Purchase Probability 0.3504 (35.04%)

The trained Logistic Regression model was used to demonstrate prediction on a new customer record. At the default classification threshold, the model predicts no purchase with a purchase probability of 35.04%

6. Conclusion

This project developed a customer purchase prediction system using Logistic Regression and Random Forest on a dataset containing 5,000 customer records and 13 columns. The workflow included data inspection, duplicate removal, missing-value treatment, identifier removal, categorical encoding, numerical scaling, exploratory data analysis, unsupervised clustering, supervised model training, hyperparameter tuning and multi-metric evaluation. K-Means, Elbow Method, Silhouette Score, DBSCAN and Hierarchical Clustering were used to explore customer structure without using the purchase target. Logistic Regression and Random Forest were evaluated for purchase classification, and Random Forest hyperparameters were tuned using 5-fold GridSearchCV. The final demonstration showed how the trained Logistic Regression model can predict the purchase outcome and probability for a new customer.

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Signature of the Guide: _____ **Course Instructor:** _____